Institutional Asset Allocation meets the Efficient Frontier: Replicating the JPMorgan Efficiente 5 Strategy

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1 The Efficiente 5 Index

The JPMorgan Efficiente 5 (JPME5) index is a rules-based, global, tactical multi-asset class strategy implemented using ETFs. At its core, JPME5 is a straightforward application of modern portfolio theory (MPT). However, its use of a short-term (6 months) lookback window for input calculation also gives it a time series momentum flavor. This paper explains how the JPME5 strategy is constructed through a replication. It accompanies an open-source replication in Python.

2 Modern Portfolio Theory Primer

Modern Portfolio Theory (MPT), developed by Harry Markowitz in the 1950s, formalized the idea that risk and return must be considered jointly when constructing a portfolio. The key insight of MPT is that, when a portfolio contains multiple assets, its risk depends on how the assets move relative to each other, which is captured by their covariances. Combining imperfectly correlated assets can reduce overall volatility without necessarily reducing expected return.

In practice, MPT frames portfolio construction as an optimization problem: find the set of portfolio weights that maximize expected return for a given risk level, or equivalently, minimize risk for a given return. The resulting efficient frontier traces all such optimal portfolios. Investors choose among them depending on their risk tolerance. If we denote the return of a portfolio of n assets by r_P and its expected return and variance, respectively, by $E(r_P)$ and $\text{Var}(r_P)$, we can write the Markowitz problem mathematically as follows:

$$\min_{w_1,\ldots,w_n} \text{Var}(r_p) \quad \text{such that} \quad \sum_{i=1}^{n} w_i = 1 \quad \text{and} \quad E(r_p) = \overline{r} \tag{1}$$

In words: find the portfolio weights $w_1, w_2, \ldots, w_n$ (proportions to invest in each asset) that minimize the variance of the portfolio, subject to the constraint that all the capital must be invested in the portfolio (the weights must sum to 100%) and that the investor requires a target expected return, $\overline{r}$.

In his seminal paper, Markowitz (1952) was careful to state that the application of MPT requires suitable capital market assumptions about risk and return, which are needed to calculate the portfolio's expected return and variance in equation (1). In practice, naïve applications of MPT, which rely on sample moments calculated from historical returns, are plagued by a variety of problems, which typically result in overly concentrated portfolios and unstable portfolio weights over time (Michaud, 1989; Best and Grauer, 1991). A common "trick" to rein in these issues is the introduction of constraints. For example, adding a no-short sale constraint, which implies that portfolio weights cannot be negative ($w_i \geq 0$), not only makes practical sense to most investors, but also significantly alleviates some of the common issues associated with MPT in practice (Jagannathan and Ma, 2003). As we will see next, the JPME5 strategy introduces several constraints that can help to "discipline" MPT optimal weights, even when relying on historical data over a short period of time to calculate the inputs.

3 Theory meets Practice: The JPMorgan Efficiente 5 Strategy

The JPME5 strategy is a direct application of the MPT framework described above, where individual assets are the ETFs which are grouped by asset class as shown in Table 1.[1] In addition to listing the individual constituents of the strategy, the table also provides caps that indicate the maximum allowed exposure by asset class and individual assets.[2] For example, the exposure to the "Developed Equity" asset class is capped at a maximum of 50% of the portfolio. This asset class contains three ETFs, which correspond to U.S. equities (SPY), U.S. small caps (IWM), and developed markets equities (excluding the U.S.). In turn, the exposure to each of these ETFs is capped at the values shown in the table (20%, 10%, and 20%, respectively).

Mathematically, we should note that the introduction of individual asset class constraints results in a modification of the optimization problem (1) through the use of inequalities to reflect minimum and maximum weights per asset and asset class/sector. The problem remains within the class of convex optimization problems, which are well understood and for which computationally efficient routines exist.[3] As we discuss later, the official documentation of the strategy does not discuss covariance matrix estimation or even optimization, mentioning instead that the strategy works by inspection of multiple hypothetical portfolios. In our replication, we use the traditional MPT approach, obtaining efficient portfolios by solving the corresponding optimization problem.

In MPT applications, it is common to use 3 to 5 years of daily data to

[1] The strategy maintained by JPMorgan considers the non-investable index JPCAUS3M (JPMorgan Cash Index USD 3 Month) as the proxy for cash. To keep all assets investable, we replace it with the BIL ETF.

[2] This is taken directly from the documentation provided by the index provider, available here, which also contains the full name of each ETF. Note that JPMorgan refers to asset class as "sector". We prefer the former, as it better reflects what the groups are meant to represent.

[3] A short explanation is provided in the documentation of the `pypfopt` Python package, which we use in our replication.

Table 1: Constituents of JPMorgan Efficiente 5

Asset class (cap)	Asset	Ticker	Asset Cap
Developed Equity (50%)	U.S. Equities	SPY	20%
	U.S. Small Cap Equities	IWM	10%
	Dev. Market Equities (ex U.S)	EFA	20%
Bonds (50%)	Treasuries	TLT	20%
	Investment Grade Bonds	LQD	20%
	High Yield Bonds	HYG	20%
Emerging Markets (25%)	Emerging Market Equities	EEM	20%
	Emerging Market Bonds	EMB	20%
Alternative Investments (25%)	Real Estate	IYR	20%
	Broad Commodities	GSG	10%
	Gold	GLD	10%
Inflation Protected Bonds/Cash (50%)	Inflation Protected Bonds	TIP	50%
	Cash	BIL	50%

construct optimization inputs. The reason is to ensure that the estimates are reliable, particularly when the number of assets is large, which however is not the case as the strategy uses only 13 ETFs. As mentioned previously, the use of a relatively short lookback window of 6 months is meant to exploit asset/asset class momentum, as hinted at in the JPME5 strategy documentation.

The portfolio construction rules are as follows:

1. The portfolio is rebalanced every month.

2. At the end of each month, the average return and covariance matrix of the ETFs are calculated using a lookback window of 6 months (126 days).

3. Using the average returns and covariance matrix calculated in the previous step, the efficient portfolio with the highest return with a target volatility of 5% is selected, subject to the asset class and individual asset constraints

in Table 1. If no portfolio with a volatility of 5% is feasible, the target volatility is increased by 1% and the process is repeated.[4]

4. In addition to the asset class and asset constraints, the minimum weight for any asset must be a multiple of 5%.

We also note that the final result should be an *excess return index*, i.e. the portfolio is funded by a short position in cash (which we proxy for using the BIL ETF). Therefore, in our backtests, we subtract the daily returns of BIL from the (fully funded) strategy returns.

4 Replicating the JPMorgan Efficiente 5 strategy

The replication of the JPME5 strategy in this paper (and the accompanying code) aims to provide a standalone, easy-to-understand working example of what backtesting a systematic tactical allocation strategy entails. The code can be run by anyone with a basic knowledge of Python. We have kept the code as simple as possible while aiming to obtain a realistic backtest of the replication.[5] We rely on Yahoo Finance for data extraction and well-known Python libraries to do the heavy lifting.

We obtain data for all the ETFs in Table 1 from Yahoo Finance over the period from January 2008 to August 2025. For comparison purposes, we also obtain total return data for the JPME5 index over the same period. Since the strategy uses 126 trading days to calculate inputs, our backtested results are for the period from August 2008 to August 2025 (a total of 17 years).

The top left panel of Figure 1 shows a scatter plot of the daily returns of the actual JPME5 strategy versus the daily returns of our replication. We also

[4] There may be instances in which the lowest volatility attainable, i.e. the volatility of the minimum variance portfolio, is higher than the target volatility. In such cases, we use the minimum variance portfolio weights.

[5] Keeping code complexity and abstraction to a minimum comes at the cost of lower code reusability/generality.

Table 2: Return statistics

	JPME5 (replication)	JPME5 (actual)
Average Excess Return	3.49%	2.89%
Volatility	6.23%	6.07%
Sharpe	0.58	0.50
Sortino	0.79	0.68
Max. Drawdown	-19.07%	-16.56%
Avg. Monthly Turnover	42.48%	-

overlay a regression line. The results suggest that, overall, there is a good fit between the replication and the index, although we note that a few daily returns deviate significantly from the regression line. The correlation between the daily returns of our replication and those of the index is approximately 0.94. The top right panel shows the cumulative excess returns of our replication (blue) and the actual JPME5 index (orange). This graph confirms that the replication tracks the index quite well. In fact, our replication outperforms the actual strategy slightly.

In terms of drawdowns, the bottom left panel of Figure 1 shows that these are largely similar, with the replication showing slightly lower drawdown during late 2015, and slightly higher drawdown during the COVID turmoil in early 2020. These results suggest that the larger differences between the replication and the actual strategy are confined to a few specific periods. To investigate this, in the bottom right panel of the figure, we plot the 63-day rolling tracking error of the replication.[6] With a few exceptions, the tracking error is quite low (the average and median trackng errors are 1.48% and 1.10%, respectively). The largest tracking errors are observed during the 2008 crisis and the 2020 COVID period.

Table 2 shows return statistics of the replicated strategy and the JPME5

[6] We define the tracking error as the annualized standard deviation of the difference between the daily return of the replication and that of the actual index.

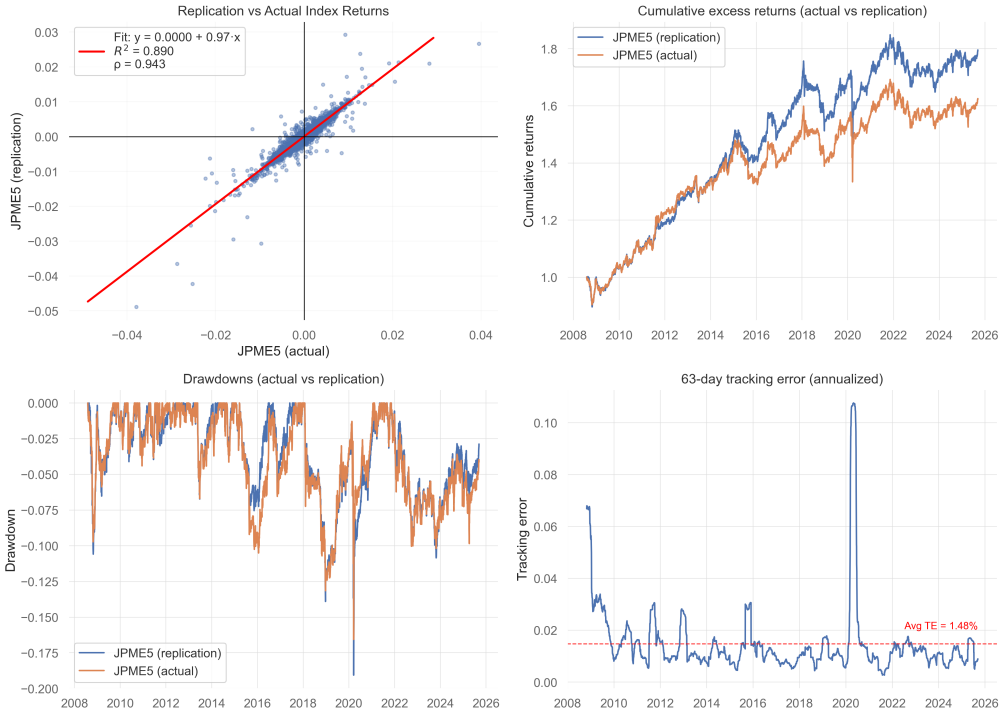


Figure 1: Cumulative excess returns (top) and average allocation weights per year

index. As expected from the previous results, the replication slightly outperforms the actual index, both in terms of average returns and on a risk-adjusted basis as measured by the Sharpe and Sortino ratios. The maximum drawdown of the replicated strategy, which corresponds to the COVID period, is higher than that of the actual index.

4.1 What explains the differences?

The differences between our replication and the published strategy are not unexpected. While our implementation follows the spirit of the strategy, as described in its documentation, a number of modelling choices can explain why the results ocasionally diverge:

- **Calculation of inputs** - in our implementation, we used the average compounded returns and the sample covariance matrix over the last 126 trading says. This is meant to reflect the performance over the last 6 months, as described in the strategy documentation. It is possible that the actual strategy uses a different approach to calculate the inputs. We have tried variations using the daily returns in the the last 6 calendar months, and using simple (instead of geometric) average to calculate the performance of the individual assets and obtained similar results to those reported in this paper.

- **Optimization approach** - we have relied on standard MPT optimization with an additional step to discretize the weights to multiples of 0.05. While the strategy documentation[7] mentions MPT and the concept of the efficient frontier, it does not directly mention optimization. Instead, it states that "The strategy assigns the weights to the Basket Constituents based upon the returns and volatilities of multiple hypothetical portfolios comprising the Basket Constituents measured over the previous six months." This suggests that the strategy relies on a search approach to obtain the optimal portfolio, which likely explains occasional differences.

[7] See the Index Supplement J.P. Morgan ETF Efficiente 5 Index

- **Other reasons** - another source of potential differences between the replication and the original strategy may be our use of the BIL ETF as a proxy for cash, instead of the index used by JPMorgan.

4.2 Variations of the base strategy

As a rules-based strategy, JPME5 uses fixed choices for key parameters that determine allocation, such as the 6-month lookback window, the monthly rebalancing frequency, and the 5% target volatility. An interesting question is to which extent the performance of the strategy depends on these choices. To investigate this, we perform backtests of our version of the strategy using the following grid of values:

- Rebalancing frequency: weekly or monthly
- Lookback window: 6 months (126 days) or 12 months (252 days)
- Target volatility: 5% or 10%

Table 3 reports statistics for the variations of the base strategy. Numbers in bold in each row denote the variation with the best result for the corresponding metric. Panel A reports results for the baseline target volatility of 5%. The results suggest that using a weekly rebalancing frequency leads to a deterioration in average as well as risk-adjusted returns, relative to monthly rebalancing. At the same time, the more frequent rebalancing appears to reduce the maximum drawdown. In the case of weekly rebalancing, the length of the lookback window (6 or 12 monhts) does not appear to affect the results much. For monthly rebalancing, however, using a lookback of 6 months produces generally better results. Overall, these results suggest that the strategy is more sensitive to the rebalancing frequency than the lookback window. The performance of the strategy with 5% target volatility is optimal with the choices for the rebalancing frequency and lookback window of the published strategy.

Panel B of Table 3 reports results with a target volatility of 10%, which is arguably more relevant for institutional investors. The realized volatility of

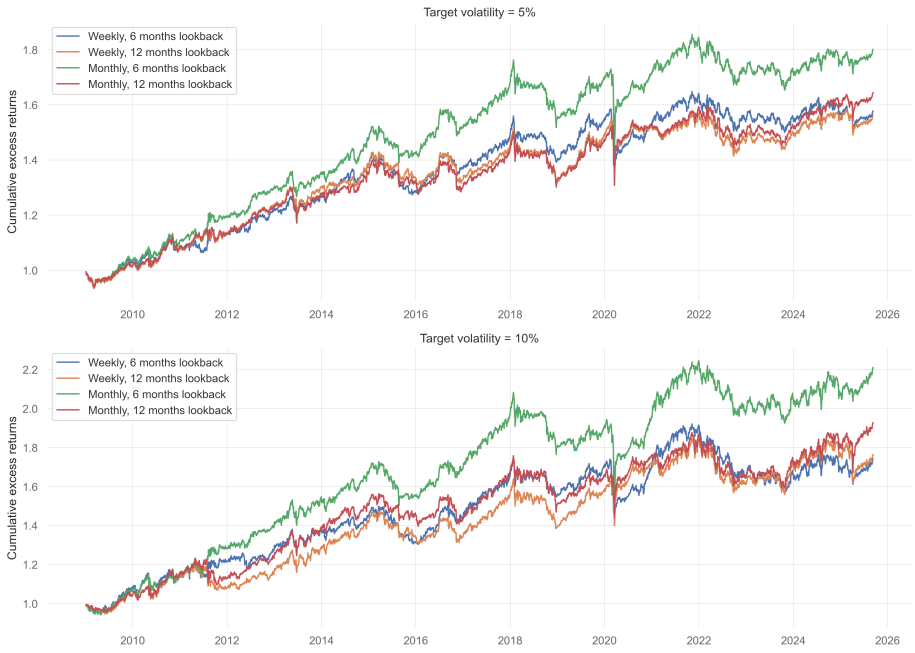


Figure 2: Cumulative excess return of variations

all variations of the strategy are significantly below the target volatility. This suggests that the individual asset and asset class constraints are binding and the constrained efficient frontier does not reach the desired volatility. The risk-adjusted returns are quite similar to those in Panel A, with higher drawdowns as expected. As in Panel A, the combination of monthly rebalancing with a lookback window of 6 months produces better results than other variations. We conclude that, to reliably reach a higher target volatility, one would need to relax the constraints in Table 1.

Figure 2 shows the cumulative excess return of the variations of the base strategy for target volatilities of 5% (top panel) and 10% (bottom panel). Confirming our results from Table 3, in both cases, monthly rebalancing using a lookback window of 6 months appears to dominate other variations.

Table 3: Variations of the base strategy

Panel A: Target volatility = 5%

	Rebalancing frequency			
	Weekly		Monthly	
Lookback window	6 months	12 months	6 months	12 months
Average Excess Return	2.90%	2.83%	**3.72%**	3.15%
Volatility	5.58%	**5.48%**	6.15%	5.76%
Sharpe	0.52	0.52	**0.61**	0.55
Sortino	0.70	0.69	**0.82**	0.75
Max. Drawdown	-15.01%	**-13.73%**	-19.07%	-14.19%
Avg. Turnover	21.75%	**16.33%**	42.49%	30.70%

Panel B: Target volatility = 10%

	Rebalancing frequency			
	Weekly		Monthly	
Lookback window	6 months	12 months	6 months	12 months
Average Excess Return	3.65%	3.74%	**5.12%**	4.31%
Volatility	**7.99%**	8.16%	8.43%	8.56%
Sharpe	0.46	0.46	**0.61**	0.50
Sortino	0.62	0.62	**0.83**	0.69
Max. Drawdown	-19.39%	**-15.52%**	-25.71%	-18.13%
Avg. Turnover	27.40%	**19.31%**	49.51%	33.80%

Conclusion

This paper discusses a replication of the JPMorgan Efficiente 5 (JPME5) strategy, a systematic, multi-asset class strategy implemented with ETFs. Our independent results are very close to the original JPME5 strategy, with a correlation of 0.94 with the published returns, with occasional divergences in moments of higher market volatility explained by potential differences in implementation choices. The accompanying code allows any interested reader with a minimum knowledge of python to verify our implementation.

References

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